

Classical electromagnetic and gravitational field theories as linear irreversible thermodynamics (LIT) of a vortex sponge

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This note develops the construction that followed the record of 15 September 2026 [37]. That note is not modified. A symbol or named concept is defined at first use, or is pointed to the section where it is defined. Flux density means the gravitational flux density $\Phi^{\lambda\mu\nu}$ (rank 3) or the electric flux density $\mathbf{E}$ (rank 1; usually called the electric field). Current is reserved for the electric current $\mathbf{j}$ (and j^μ) and the energy current T^{0i} (and $T^{\mu\nu}$). Whenever the three constitutive descriptors of the vortex sponge are named together they are written in this order: inertial density, flow, tension tensor (ρ, V, Θ_{ij}) . Companion records: constitutive dictionary (20 August 2026) [34]; electric charge as conversion [36]; rank-2 potential and rank-3 flux [37]; philosophical foundations [35].

Abstract

Electromagnetism and gravity are two aspects of one medium, the vortex sponge. Both are treated as linear irreversible thermodynamics of that medium.

Electromagnetism is the coupling of durable conversion sites (positive and negative charges) to the fields they sustain. A charge distribution sets gradients of the potentials ϕ and $\mathbf{A}$. Those gradients drive the electric flux density $\mathbf{E}$ and the magnetic induction $\mathbf{B}$. The flux densities react back on the charges. The flux lines carry inertia, so a time-dependent pattern can oscillate; that oscillation is an electromagnetic wave.

Gravity is the coupling of durable dissipative structures (masses, concentrations of energy) to the constitutive state of the same sponge. Those structures set gradients of a rank-2 potential $\Psi^{\mu\nu} = (\Psi^{00}, \Psi^{0i}, \Psi^{ij})$. The components of Ψ map one-to-one onto the coarse descriptors of the sponge: inertial density, flow, and tension (ρ, V, Θ_{ij}) . The gradients drive a rank-3 gravitational flux density $\Phi^{\lambda\mu\nu}$. Because $\Phi^{\lambda\mu\nu}$ carries a stress, it sources a further increment of $\Psi^{\mu\nu}$, which drives a further increment of $\Phi^{\lambda\mu\nu}$, and so on: the field is obtained by iteration. The same three descriptors (ρ, V, Θ_{ij}) induce the effective metric felt by every excitation of the sponge.

The note opens with nested levels of reality, then the thermodynamics, then the one-medium statement. The static rank-2 potential and rank-3 flux density of the 15 September note [37] are the special case with which the field equations begin.

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1 Nested levels of reality

This note starts from a statement about levels of description, not from a list of fields [35]. Ordinary laboratory matter, and the fields that couple to it, are treated as excitations of an underlying continuum — a substrate, or medium. That statement belongs to a hierarchy of levels of analysis. The same relation is assumed to hold again above the substrate and again below ordinary matter: what one description sees as a conserved object is, at a finer level of analysis, a process in a finer medium, and each level is described with the continuum mechanics appropriate to that level [26, 33, 29]. Internal and external observers, and the stock and rate readings of one and the same accounting, are two levels of analysis of that statement. They are introduced in the subsections below. The constitutive potential, the gravitational flux density, and the source tensor of the later sections are named when those sections begin (Sections 4 and 5).

1.1 Nested levels

Written as a hierarchy of nested levels of reality, the relation is this [26, 33, 29, 35]. A coarser world takes the conversion regions of a finer world as its conserved objects, and takes its own excitations as objects for a world coarser still:

$$\text{PRIME REALITY} \rightarrow \text{substrate} \rightarrow \text{ordinary matter} \rightarrow \text{analogue / acoustic matter}.$$

Analogue gravity is the last arrow written as laboratory physics: a fluid or an optical medium produces an effective metric for its own excitations [29, 33]. The present note is the same statement at the next level of analysis: ordinary matter and its fields are excitations of the substrate, and they move on the effective metric built from the constitutive potential Ψ of the substrate (introduced in Section 4; the metric map is Section 9).

The methodological rule is that each level is described with the continuum mechanics appropriate to that level [35]. The substrate, at the scale of elementary localized sources, is treated as a continuum with torsional stiffness Θ and inertia density ρ (Section 3). Granularity of the substrate, if it is needed, is assigned to a much smaller scale (the grain whose crossing time is the Planck time t_P , Section 6).

1.2 Nature repeating the same pattern

A vortex is, for a two-dimensional dweller on a water surface, what a particle is for us. That dweller is himself embedded in the surface: he is a pattern on the same water [4, 30]. A chemical reactor is, for a description that only considers concentrations, what a conversion region is for a description that considers affinities and rates [15, 28]. The electromagnetic dipole is a pair of conversion regions of opposite function [36].

The pattern is the same. Only the tensor rank and the conservation law change.

1.3 Internal observer and external observer

An observer made of the same excitations as the objects under study — an internal, intra-substrate observer — necessarily considers stocks. A stone is a stone, a charge is a charge, energy is conserved. That is the laboratory description: the internal description.

The same contrast is the standard one between dwellers inside a fishbowl universe and the laboratory that holds the medium [30, 33, 29]: internal observers, built from the excitations, reconstruct an effective relativistic world and cannot, by operations available inside that world, detect the laboratory frame of the substrate. They are patterns in the medium, as the surface dweller is a pattern on the water. The older image of the same procedure is the cave of *Republic VII* [25]: the dwellers see the shadows; the external account sees what casts them. An observer of the medium itself — an external observer, the “divine” eye relative to the etheric layer — considers rates. The primitives are the degrees of freedom of the substrate. What the internal observer calls mass or charge is a conversion region that happens to be durable and coherent. The analogy is exact with a

vortex in water, or with a chemical reactor: from the surface of the water the vortex is an object, and the dweller is another object of the same kind; from the laboratory both are flow patterns [4, 28].

In the language used elsewhere in this program, the upper level of that hierarchy is written PRIME REALITY [35]. The word marks the next description, not a physical field in the present equations.

1.4 Stock and rate

In the laboratory description, the energy–momentum tensor $T^{\mu\nu}$ of ordinary matter (Section 5) is a conserved energy–momentum density [15, 24, 37]. The balance law looks like conservation of that density. Mass and charge are properties of objects. Poisson’s equation is written in these variables.

In the description of the medium, the same tensor is a conversion-rate tensor $\Pi^{\mu\nu}$ multiplied by the Planck time $t_P = \sqrt{\hbar G/c^5}$ (the working value of the conversion factor τ_* in Section 6),

$$T^{\mu\nu} = \Pi^{\mu\nu} t_P. \quad (1)$$

Mass and charge are then long-lived conversion processes: durable conversion sites in the vortex sponge, or steady conversion between two allotropes of the medium [36, 23, 28, 9]. Joule’s law is written in these variables.

The two readings differ by a time. They do not differ by a mechanism.

The identification $\tau_* = t_P$ is a working hypothesis, not a derived result. It is provisional: another characteristic time of the substrate could stand in the same slot, and the laboratory alternative $\tau_* = L/c_*$ is left open in Section 6. What the hypothesis attempts to say is only this. The intra-substrate description sees lasting objects (stocks). The extra-substrate description sees a steady-state flow. The map between those two accounts is a time, and the trial value of that time is t_P , so that $1/t_P$ is being tried as the characteristic sampling rate of the host description — a frame rate of the universe, in that restricted sense. The image is the cave already cited, or a cinema [25]: the figures on the screen are, for the dwellers, persistent objects; in the host hall they are a steady stream of light and colour. The intra-substrate account sees the figures. The extra-substrate account sees the stream. The time τ_* converts the stream into the figures. It does not add a second substance.

That accounting is implemented by linear irreversible thermodynamics, which is the next section.

2 Linear irreversible thermodynamics

The structure used above is not an analogy invented for this program. It is linear irreversible thermodynamics, also called thermodynamics of irreversible processes [16, 17, 20, 24, 23]. Entropy production is a sum of products of fluxes and forces,

$$\sigma_s = \sum_{\alpha} J_{\alpha} X_{\alpha}, \quad J_{\alpha} = \sum_{\beta} L_{\alpha\beta} X_{\beta}, \quad (2)$$

with the Onsager–Casimir reciprocity $L_{\alpha\beta} = \pm L_{\beta\alpha}$ [16, 20]. The even part of $L_{\alpha\beta}$ is dissipative. The odd part (Casimir) is gyrotropic: it couples fluxes without producing entropy.

The standard textbook is de Groot and Mazur [24]. Onsager’s two papers fix the reciprocity for time-even forces [16, 17]. Casimir extends the reciprocity to magnetic and Coriolis-type forces [20]. Prigogine’s monograph states the linear theory and the theorem on minimum entropy production in the stationary state [23]. That theorem is a corollary of the linear theory, not a second framework.

Applied to the medium itself, the same skeleton says: the potentials are intensive constitutive fields of the substrate; the field strengths are the fluxes those fields drive; the sources are conversion regions. Electrodynamics and the rank-2 gravitational construction of Sections 4–5 occupy two blocks of one matrix $L_{\alpha\beta}$.

This identification is the working hypothesis of the note. It does not, by itself, produce Faraday kinematics, a finite wave speed, or an effective metric. Those are additional constitutive statements, developed below.

That structure is now applied to one medium. Electromagnetism and gravity are two aspects of the same vortex sponge (Section 3), not two ethers.

3 One medium

A working hypothesis of this research program is that the fields of laboratory physics — the strong and weak interactions, electromagnetism, and gravity — are aspects, modes of motion, and excitations of a single underlying substance [4, 9, 8, 7, 35]. That substance may be called substrate, medium, aether, or pneuma; the name is not the claim. The program is then to identify the material properties of the substance and to show how its dynamics produce the sources, potentials, flux densities, and effects of each interaction. The present note gives the elements of that identification for the two classical field theories, electromagnetism and gravitation [3, 22, 34, 36, 37]. Because both theories are written on one medium, they cannot live in two different continua with two different characteristic speeds.

3.1 Micropolar microstructure

The candidate medium is a vortex sponge in the compact-ring realization [4, 9, 8, 7]. It is not a connected web of long vortex filaments.

It is an ideal fluid seeded with compact, randomly oriented vortex rings. Neighbouring rings couple through the potential flow they induce in the interstitial fluid: mutual hydrodynamic induction. No ring has to span from one coarse cell to the next.

After coarse graining this is a micropolar fluid in Eringen’s sense, equivalently a Cosserat fluid [5, 6]. Each cell has a translational velocity V and an independent microrotation. The cells exchange force and torque. The torque per unit area is the couple stress. The coarse torsional stiffness of that coupling is the tension Θ_{ij} . The inertia of the interstitial fluid is ρ . The ordered constitutive triple is (ρ, V, Θ_{ij}) .

Because translation and microrotation are independent fields, a standing differential shear of V does not destroy the carrier of torsion. Adjacent groupings of rings may drift at different coarse speeds — a Kerr-like stack of rings. They still torque one another through the local induced flow. A linear torsional wave is a small extra microrotation that hops from group to group. It does not ride a filament that joins the two groups, so it does not care about their relative bulk velocity except as a local Fizeau drag. That wave is the electromagnetic field of later sections.

Transfer of energy from the rotational structures to the interstitial fluid lowers Θ and raises ρ ; the wave speed falls [9].

3.2 One characteristic speed

Electromagnetic fields are excitations of this sponge. The local wave speed is not a posted constant. If c_0 is the far-field value of $\sqrt{\Theta/\rho}$ and $\bar{n}$ is the isotropic refractive index of the optical dictionary (Section 9), the local speed is

$$c_*^2(x) = \frac{\Theta(x)}{\rho(x)} = \frac{c_0^2}{\bar{n}(x)^2}. \quad (3)$$

Gravitational disturbances, being disturbances of the same constitutive pair, travel at the same speed. The coefficients κ_{EM} and κ_{G} are the two conductivities, one for the electric flux density and one for the gravitational flux density; $\kappa_{\text{G}} = c^2/(4\pi G)$ is fixed in Section 5. They fix how much flux density a given source produces. They do not fix two light cones.

3.3 Impedance matching

Impedance matching [34],

$$Z = \rho (\det \Theta)^{1/3} = Z_0, \quad (4)$$

is the statement that the geometric mean of inertia and stiffness is uniform. It leaves one isotropic degree of freedom, $\bar{n}$ or c_* . That degree of freedom is the time-time slot g_{00} of the effective metric (Section 9). The electromagnetic ratio μ/ε is not imposed here. It is derived from one species of charged rotor in Section 7.6.

3.4 Gravity as a standing constitutive bias

Gravity is therefore not a second fluid. It is a slow, standing change of (ρ, Θ) , maintained by conversion regions and by dissipation of the gravitational flux density Φ (Section 4).

The sign of the change is fixed by three already locked statements: the optical metric of the dictionary, Kelvin's drop of wave speed with vortex intensity, and the converter picture of a localized source [9, 36, 34]. A gravitational well has $\bar{n} > 1$ and $c_* < c_0$. At constant Z_0 this means that Θ decreases and ρ increases. In the far field the complementary state holds: larger Θ , smaller ρ , $c_* = c_0$. A localized energy concentration is a sink of torsional stiffness and a source of inertia. The static potential profile of $\bar{n}$ around it is the long-range profile of that conversion.

The static construction of 15 September is the special case with which the field content begins.

4 Starting point

The 15 September note [37] locked a static, linear theory in three-dimensional Euclidean space. A symmetric potential Ψ^{ij} defines a gravitational flux density by Fick's law, $\Phi^{ijk} = -\kappa \partial_k \Psi^{ij}$. A symmetric density σ^{ij} sources that flux density. Throughout these notes Φ is the gravitational flux density and $\mathbf{E}$ is the electric flux density (usually called the electric field). The word *current* is reserved for the electric current $\mathbf{j}$ (and j^μ) and for the energy current T^{0i} (and $T^{\mu\nu}$). With σ^{ij} identified as matter stress and Ψ^{ij} as a gravitational potential, the interaction density is $\sigma : \Psi / c^2$ and Poisson's equation sits in $\kappa = c^2 / (4\pi G)$. In the fluid reading, σ^{ij} is a conversion region, not a third fluid.

Three questions were left open, and they organize what follows. First: how does this static theory become a 3+1 theory of gravity, with a scalar, a vector and a tensor assembled into one $\Psi^{\mu\nu}$, and with the matter energy-momentum tensor as the seed source? Second: why does the flux density carry a stress, and why must that stress be fed back into the source? Third: how do localized energy concentrations, which are themselves excitations of the medium, come to move on an effective metric built from Ψ ?

The answers below stay inside linear irreversible thermodynamics [16, 17, 20, 24, 23]. They do not replace the 15 September equations. They say what those equations are a special case of.

5 Field content and balance laws

The background remains Euclidean three-space with Newtonian time. These are the coordinates in which the medium is described. They are not the geometry measured by matter.

5.1 Rank-2 potential and the constitutive pair

Once time is restored, the intensive state of the medium has three mechanical sectors: an isotropic inertia-stiffness slot, a local flow, and a stiffness tensor. They are not three independent theories. They are the components of one symmetric tensor $\Psi^{\mu\nu}$,

$$\Psi^{00} \text{ isotropic slot}, \quad \Psi^{0i} = V^i \text{ (flow)}, \quad \Psi^{ij} \text{ stiffness ellipsoid.} \quad (5)$$

The map onto the mechanical fields of the vortex sponge is one-to-one, at fixed impedance [34]: Ψ^{00} tracks the isotropic pair (ρ, Θ) (equivalently $\bar{n}$ or c_*); Ψ^{0i} is the local flow V^i of the sponge; Ψ^{ij} tracks the stiffness ellipsoid Θ^{ij} . Whenever those three descriptors are named together they are written in the order (ρ, V, Θ_{ij}) : inertial density, flow, tension tensor. The 15 September field Ψ^{ij} is

the spatial block of this object. The identification is constitutive: Ψ is the mechanical state of the micropolar sponge of Section 3.1, not an auxiliary potential invented to integrate an equilibrium equation.

The source that starts the construction is the energy–momentum tensor $T^{\mu\nu}$ of ordinary matter — energy density, energy current T^{0i} , and stress of localized energy concentrations [36, 37]. This $T^{\mu\nu}$ is the seed. It is not yet the full source of Ψ .

5.2 Gravitational flux density: tensorial Fick law

A gradient of Ψ drives a gravitational flux density of the same tensor type, with one extra index for the direction of transport [2, 16, 24],

$$\Phi^{\lambda\mu\nu} = -\kappa \partial^\lambda \Psi^{\mu\nu}. \quad (6)$$

This is Fick’s law, or Ohm’s law, written for a rank-2 potential. The coefficient κ is a transport coefficient of the medium. The value that reproduces the Newtonian Poisson equation on the spatial block is [1, 31]

$$\kappa = \frac{c^2}{4\pi G}. \quad (7)$$

Large κ (small G) means that a small gradient of Ψ already carries the flux density demanded by the source.

5.3 Self-coupling through the Belinfante–Papapetrou stress

The construction is a rank-2 field theory: a potential $\Psi^{\mu\nu}$, a gravitational flux density $\Phi^{\lambda\mu\nu}$, and a rank-2 source. In the internal, intra-etheric description that source is identified with the energy–momentum tensor. Whatever carries an energy–momentum tensor is therefore a source of Ψ . Ordinary matter supplies the seed $T^{\mu\nu}$. The gravitational flux density occupies a region of space and itself carries a Belinfante–Papapetrou tensor $\theta^{\mu\nu}$ [18, 19, 21], built from the Noether stress of Φ and from the spin current of the tensor indices of Ψ . That region is therefore also a source.

In the external, rate-like description the same fact is a dissipation. Wherever a gravitational potential gradient drives a gravitational flux density, the product $(\nabla\Psi) : \Phi$ is the local conversion rate: organized ring-vortex structure is converted into unstructured fluid (Section 6). That conversion is booked as $\theta^{\mu\nu}$ after the map by τ_* . It sources more Φ , which sources more conversion, and so on.

The balance law is therefore

$$\partial_\lambda \Phi^{\lambda\mu\nu} = T^{\mu\nu} + \theta^{\mu\nu}. \quad (8)$$

On the flat background, in the harmonic gauge $\partial_\lambda \Psi^{\lambda\nu} = 0$, this is Sciama’s equation in gothic form [22, 27],

$$-\kappa \square \Psi^{\mu\nu} = T^{\mu\nu} + \theta^{\mu\nu}. \quad (9)$$

5.4 Iteration

The field is obtained by iteration: compute Φ from the seed T , compute θ from that Φ , replace the source by $T + \theta$, and repeat until the flux and the source agree. The static field of the seed alone is only the first step of that loop.

Matter does not propagate with the Euclidean background metric. It propagates with the effective metric built from Ψ (Section 9). That statement is a constitutive law for the excitations of the medium, not a consequence of the balance law alone.

6 Why the flux density carries a stress

The 15 September note associated a Maxwell-like stress with Φ because a scalar energy density cannot be added to a tensor source. The material reason for that stress was not yet given.

In linear irreversible thermodynamics the local dissipation is the product of the flux density and the force that drives it [16, 24],

$$\pi = -\Phi^{\lambda\mu\nu}\partial_\lambda\Psi_{\mu\nu}. \quad (10)$$

The units of π are W m^{-3} . Fick's law turns the product into a square,

$$\pi = \frac{1}{\kappa}\Phi^{\lambda\mu\nu}\Phi_{\lambda\mu\nu}. \quad (11)$$

This is Joule's law for the gravitational flux density. If $\kappa > 0$, then $\pi \geq 0$.

6.1 Why one number is not the whole object

The quantity π is a *contraction*. It is the full pairing of every index of Φ against every index of $\nabla\Psi$ (or, after Fick, of Φ against itself). The same factors, with two indices left free, are a symmetric tensor $\theta^{\mu\nu}$ of Maxwell–Belinfante type [3, 18, 21]. The exact split is the canonical (orbital) piece plus the Belinfante superpotential [18, 19, 21],

$$\theta^{\mu\nu} = t^{\mu\nu} + \partial_\lambda K^{\mu\nu\lambda}, \quad (12)$$

$$t^{\mu\nu} = \frac{1}{\kappa}\left(\Phi^{\mu\alpha\beta}\Phi^\nu_{\alpha\beta} - \frac{1}{2}\eta^{\mu\nu}\Phi:\Phi\right). \quad (13)$$

The superpotential is built from the spin current S of the tensor indices of Ψ ,

$$K^{\lambda\mu\nu} = \frac{1}{2}(S^{\lambda\mu\nu} + S^{\mu\lambda\nu} - S^{\nu\lambda\mu}). \quad (14)$$

No new bilinear is introduced. The scalar π is recovered from $\theta^{\mu\nu}$ by a contraction; the other slots of $\theta^{\mu\nu}$ are the leftover index structure of Φ .

A Lorentz boost of a scalar cannot produce this tensor. In the rest frame of a static, spherical well the flux density is radial, π is a single number, and $\theta^{0i} = 0$, but θ^{ij} is still anisotropic: tension along the radial tubes and pressure across them, as in electrostatics [3]. That anisotropy is already present before any boost.

6.2 Rate and stock as two levels of analysis

The two levels of analysis were introduced in Section 1. Here they are applied to the flux-density bilinear.

The same tensor is read as a rate or as a stock [15, 24],

$$\theta^{\mu\nu} = \Pi^{\mu\nu}\tau_*. \quad (15)$$

The working choice is

$$\tau_* = t_P = \sqrt{\hbar G/c^5}, \quad (16)$$

the crossing time of one grain of the medium. Poisson's equation remains classical: $\hbar$ does not enter $\kappa = c^2/(4\pi G)$. It enters only this map.

An internal observer, made of the same excitations, considers stocks: energy density, momentum density, stress. An extra-etheric observer considers rates: how fast the substrate is being converted, how that conversion is transported, how it stresses the sponge. The two accounts differ by τ_* . They do not differ by a mechanism.

The 15 September field energy $u_\Phi = \Phi:\Phi/(2\kappa)$ is the stock reading of the same square that Joule's law writes as π . As a conductivity, κ produces the rate π . As a permittivity, $\varepsilon = \kappa\tau_*$ stores the density u . The static theory looks conservative because the rate has been converted into a stock.

6.3 The three slots, extra-etheric reading

Units of $\Pi^{\mu\nu}$ are those of $\theta^{\mu\nu}$ divided by time.

Π^{00} : **power density**. This is the one number already written as π , up to the conventional 1/2 that distinguishes $\Phi : \Phi / \kappa$ from $\Phi : \Phi / (2\kappa)$. It is the local power of the gravitational flux density: how much conversion is being booked, per unit volume and per unit time, at that event. In a static well of Φ , on the present LIT reading, Π^{00} is not zero: the standing flux density continues to convert. The internal observer calls the corresponding stock θ^{00} the energy density of the field.

Π^{0i} : **time rate of momentum density**. This slot belongs to the extra-etheric account. Its units are those of a force density, N m^{-3} : momentum per unit volume, per unit time. It is not a power flux and it is not a momentum density. Those two readings belong to the stock $\theta^{0i} = \Pi^{0i} \tau_*$, which is what an internal observer calls the momentum density of the web. Π^{0i} is sourced by a circulating or time-dependent pattern: a Coriolis field $\mathbf{\Omega} = \nabla \times \mathbf{V}$, a changing Φ , a moving source. In the rest frame of a static, non-rotating well, $\Pi^{0i} = 0$.

Π^{ij} : **anisotropic conversion (stress rate)**. The gravitational flux density has an axis. Along a tube of the flux density there is tension; across the tubes there is pressure [3]. Π^{ij} is the rate at which that momentum crosses a unit area. Its isotropic part (the 3-trace) tracks the same conversion as Π^{00} , with the sign fixed by the 3+1 split and by the superpotential. Its traceless part is shear: the ellipsoid of Ψ^{ij} being biased by the standing flux density. The internal observer calls θ^{ij} the Maxwell / Belinfante stress.

The three slots are one tensor. They are not three independent dissipations.

6.4 The same tensor as a source of more Ψ

The balance law is

$$\partial_\lambda \Phi^{\lambda\mu\nu} = T^{\mu\nu} + \theta^{\mu\nu}. \quad (17)$$

Every slot of $\theta^{\mu\nu}$ is a source for the matching slot of $\Psi^{\mu\nu}$: θ^{00} biases the isotropic constitutive field (clocks, $\bar{n}$); θ^{0i} biases the flow V^i ; θ^{ij} biases the stiffness ellipsoid. That is the second job of the same object. Self-coupling is the statement that the conversion booked by θ drives more gravitational flux density.

The seed $T^{\mu\nu}$ is the same kind of object, localized on compact conversion regions (ordinary matter). The field term $\theta^{\mu\nu}$ is the distributed conversion of the standing flux density. There are not two ontologies, only two supports.

Radiation is a different mechanism (Section 8). The laboratory alternative $\tau_* = L/c_*$, with L a size of the conversion region, is left open.

7 Gradient flux density and induction

Fick's law says what the flux density is when the source is static. It does not say how a flux density that moves, or a flux density that changes in time, is seen from the laboratory.

7.1 Two tensors built from Ψ

The gravitational flux density is the gradient,

$$\Phi^{\lambda\mu\nu} = -\kappa \partial^\lambda \Psi^{\mu\nu}. \quad (18)$$

The integrable field strength is the antisymmetrized derivative [32],

$$F^{\lambda\mu\nu} = \partial^\lambda \Psi^{\mu\nu} - \partial^\mu \Psi^{\lambda\nu}. \quad (19)$$

Electrodynamics uses both: Faraday's $F = dA$ is kinematics; $\mathbf{E}$ is the electric flux density and $\mathbf{D}$ is the polarization state of bound opposite pairs [32, 36]. The gravitational construction of Ref. [37] used the Fick flux density Φ and the source. The Faraday tensor is needed as soon as that flux density moves or changes in time.

7.2 One-to-one comparison with electrodynamics

Maxwell theory is built from a scalar potential ϕ and a vector potential $\mathbf{A}$ [32]:

$$\mathbf{E} = -\nabla\phi - \partial_t\mathbf{A}, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (20)$$

The first line is the analogue of a Fick (gradient plus inertia) slot. The second line is the induction slot: the magnetic field is the curl of the vector potential. No further field is introduced; $\mathbf{B}$ is a derivative of $\mathbf{A}$.

The same split is made here, with two differences. The potential is a symmetric tensor $\Psi^{\mu\nu}$, so it has a scalar block, a vector block and a spatial-tensor block. The vector block is the flow of the medium,

$$V^i = \Psi^{0i}. \quad (21)$$

What $\mathbf{A}$ is for the electric flux density, V^i is for the gravitational flux density (Section 7.3) [10, 22].

The comparison, written in three-dimensional language, is

	Electrodynamics	Gravitodynamics
scalar potential	ϕ	Ψ^{00}
vector potential	$\mathbf{A}$	$V^i = \Psi^{0i}$
tensor potential	—	Ψ^{ij}
flux density	$\mathbf{E} = -\nabla\phi - \partial_t\mathbf{A}$	$\Phi^{\lambda\mu\nu} = -\kappa\partial^\lambda\Psi^{\mu\nu}$
polarization state	$\mathbf{D} = \varepsilon \mathbf{E}$ (density of $+/-$ pairs)	—
kinematic field	$\mathbf{B} = \nabla \times \mathbf{A}$ (magnetic field)	$\boldsymbol{\Omega} = \nabla \times V$ (Coriolis field)
extra kinematic field	—	$F^{ijk} = \partial^i\Psi^{jk} - \partial^j\Psi^{ik}$

$\mathbf{E}$ is the electric flux density: the field-line pattern of the electromagnetic channel, the coarse density of oriented mismatch lines of the two allotropes (α/β) [36].

The causal chain that produces $\mathbf{E}$ and $\mathbf{D}$ starts from the potential. A gradient of ϕ drives a differential flux of α and β . That flux density is $\mathbf{E}$. A converter of charge q feels a momentum-transfer rate

$$\mathbf{f} = q\mathbf{E} = -q\nabla\phi \quad (22)$$

(the static Faraday slot; the $\partial_t\mathbf{A}$ term is restored in the first line of the table). If the converter is free, there is no restoring force and it accelerates:

$$\frac{d\mathbf{p}}{dt} = q\mathbf{E}. \quad (23)$$

If the converter is bound to a partner of opposite function by the elasticity of the sponge, the same force is balanced by a spring,

$$q\mathbf{E} - k\boldsymbol{\xi} = 0, \quad (24)$$

where $\boldsymbol{\xi}$ is the displacement from the pair's rest position and k is the restore constant of that spring. The bound cell displaces and stays put.

Empty substrate is already seeded with such bound pairs and is globally neutral [36]. $\mathbf{D}$ is the aggregate descriptor of that polarized displacement. It is not a flux and is not an electric current $\mathbf{j}$. The constitutive law of free substrate is

$$\mathbf{D} = \varepsilon[\rho, V, \Theta_{ij}] \mathbf{E}, \quad \nabla \cdot \mathbf{D} = \rho_{\text{free}}, \quad (25)$$

with ε slaved to (ρ, V, Θ_{ij}) by the dictionary [34]. The displacement current $\partial_t\mathbf{D}$ is the time derivative of that pair density. The law $\mathbf{D} = \varepsilon \mathbf{E}$ is constitutive, not a second causal step: both fields are responses to the same $\nabla\phi$.

The restore constant of the seeded pairs is, as a working hypothesis of the same status as $\tau_* = t_P$ (Section 1.4), the Planck spring

$$k_P = \frac{m_P}{t_P^2} = 7.49 \times 10^{78} \text{ N m}^{-1}, \quad (26)$$

with resonance $\omega_P = 1/t_P$. For every laboratory frequency, including super-hard gamma radiation, the stretch follows $\mathbf{E}$ in phase. Dipole formation is dissipationless for all practical purposes. The first lag of $\mathbf{D}$ behind $\mathbf{E}$ is at the Planck frequency.

Gravity has no opposite-mass pairs, so the $\mathbf{D}$ row of the table is empty. The gravitational object $\Phi^{\lambda\mu\nu} = -\kappa\partial^\lambda\Psi^{\mu\nu}$ belongs with $\mathbf{E}$, as a flux density of the constitutive state, not with $\mathbf{D}$.

The gravitoelectric combination $-\partial_k\Psi^{00} - \partial_t V_k$ is the same flux-density slot written for the scalar–vector blocks, in parallel with $\mathbf{E} = -\nabla\phi - \partial_t\mathbf{A}$.

The curl that appears in the table is the ordinary Euclidean vorticity of the flow. It is written

$$\Omega^i = (\nabla \times V)^i = \varepsilon^{ijk}\partial_j V_k \quad (27)$$

and is called the Coriolis field [13, 11]. It is not a new dynamical field: it is the statement $\mathbf{B} = \nabla \times \mathbf{A}$, with $\mathbf{A}$ replaced by V . A rotating source produces a circulating V , hence a non-zero Ω . A test mass in that region sees a Coriolis acceleration [10, 13]. The GEM literature often writes the same object as a “gravitomagnetic” field B_g [12]. That name is not used here: it suggests a coupling of gravity to magnetism, which is not the construction. Ω is the vorticity of the constitutive flow, read as the field that produces a Coriolis acceleration.

Two remarks, to keep the analogy from being over-read.

First: $\Omega = \nabla \times V$ is $\mathbf{B} = \nabla \times \mathbf{A}$. It is not the Minkowski relation $\mathbf{H}' = \mathbf{H} - \mathbf{v} \times \mathbf{D}$. The latter is what a moving observer sees when the medium already carries an electric flux density. Both facts exist in electrodynamics; only the first is used in this subsection.

Second: electrodynamics has no tensor potential, so it has no analogue of F^{ijk} . That extra induction slot is the price of a rank-2 potential. The scalar-plus-vector truncation stops at Ω . The present theory does not, because Ψ^{ij} is a dynamical constitutive field.

The mechanical reading of F^{ijk} follows from the objects that make up the gravitational flux density. Those objects are not spherical parcels. They are ellipsoids: the local stiffness Ψ^{ij} (equivalently Θ^{ij}) has a principal-axis triad. A gradient of that triad is a change of orientation from one place to the next. Because the elements are non-spherical, that change of orientation is a distinguishable state of the medium; it is not a relabelling of an isotropic grain. The combination

$$F^{ijk} = \partial^i\Psi^{jk} - \partial^j\Psi^{ik}$$

is the kinematic record of that gradient. If Ψ^{ij} were pure trace (spherical grains), the antisymmetric slot would vanish and F^{ijk} would carry no independent information.

7.3 Material reading of $\mathbf{A}$ and of V

The rank-1 field $\mathbf{E}$ and the rank-3 field Φ count oriented lines crossing a unit surface [3, 36, 37].

The electric lines belong to a two-allotrope substrate [36]. α and β are the two allotropic forms of the sponge. A positive conversion cell turns α into β ; a negative conversion cell turns β into α . Mismatch tubes therefore begin on a $+$ site and end on a $-$ site.

The vector potential $\mathbf{A}$ is assembled from the oriented line elements that make up $\mathbf{E}$. It is a polar, velocity-like field that records how those elements move from one instant to the next.

Only the transverse part of that motion is physical. A displacement of a line element along its own tangent does not change the electric flux density through a fixed surface. That longitudinal sliding is a relabelling of points on a frozen drawing, and is the material content of the gauge shift $\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi$. Coulomb’s condition $\nabla \cdot \mathbf{A} = 0$ discards it at the coarse-grained level.

Each line element carries a signed strength λ according to its allotrope. An α element and a β element that execute the same motion contribute with opposite sign. An electron and a positron on the same circle, with the same angular speed, therefore cancel. The coarse field in a small cell of volume $\mathcal{V}$ is

$$\mathbf{A}(x) = \lim_{\mathcal{V} \rightarrow 0} \frac{1}{\mathcal{V}} \sum_{i \in \mathcal{V}} \lambda_i \mathbf{v}_{i\perp} \Delta\mathcal{V}_i, \quad (28)$$

where $\mathbf{v}_{i\perp} = \mathbf{v}_i - (\mathbf{v}_i \cdot \hat{\boldsymbol{\tau}}_i) \hat{\boldsymbol{\tau}}_i$ is the velocity of element i across its own tangent. The formula is a reading of $\mathbf{A}$ in the cell; distant electric currents still set that local stream through Ampère's law. The cross product $\mathbf{v} \times d\boldsymbol{\ell}$ is not used: it is axial, whereas $\mathbf{A}$ is polar.

The curl of this aggregated velocity is the magnetic field, $\mathbf{B} = \nabla \times \mathbf{A}$. A steady current in a long wire does not change the Eulerian drawing of the lines; $\mathbf{A}$ is then a standing velocity of those lines and $\mathbf{B}$ is its curl. The term $-\partial_t \mathbf{A}$ in Faraday's law is the time derivative of the curl-bearing part of $\mathbf{A}$, not a longitudinal slide. It is a change of the pattern of $\mathbf{E}$, which produces a circulating $\mathbf{E}$ even when the scalar potential is uniform. The momentum density of the same pattern is the Poynting slot $\theta^{0i} \propto (\mathbf{E} \times \mathbf{B})^i$, a component of a stress-energy tensor, not $\mathbf{A}$ itself.

The same construction is used for the gravitational channel. The flow $V^i = \Psi^{0i}$ is assembled from the oriented line elements that make up the gravitational flux density Φ . Only the translation of those elements is kept. Their non-sphericity — the orientation of the local ellipsoid — is not in V ; it sits in F^{ijk} of the previous subsection. The pair is therefore $\mathbf{A} : \mathbf{E}$ as $V : \Phi$: two webs, one construction.

7.4 Induction

A time-dependent source, or a source in motion, bends and displaces the lines of the flux density. Faraday's law and the Ampère–Maxwell law are the two statements of that fact [3, 32]:

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \partial_t \mathbf{E}. \quad (29)$$

The wave equation for $\mathbf{E}$ and $\mathbf{B}$ follows from these two equations. No relaxation time is required. Electromagnetic and gravitational radiation are induction: a changing flux density of one field is a source of the other.

7.5 Why electromagnetism has two current terms, and gravity does not

In a linear medium Maxwell writes

$$\mathbf{J}_{\text{tot}} = \sigma \mathbf{E} + \varepsilon \partial_t \mathbf{E}. \quad (30)$$

The first term is a conduction current: free conversion regions of opposite sign (free charges) drifting in $\mathbf{E}$. The second term is a displacement current. Part of that second term is inertia of the lines of the electric flux density. Part of it, in a polarizable region, is the current of bound dipoles: pairs of conversion regions of opposite function, whose separation changes when $\mathbf{E}$ changes [3, 32, 36]. The substrate, in the electromagnetic channel, supports both free charges and bound dipoles. Both terms are therefore present.

7.6 Material origin of ε and μ

The permittivity ε and the permeability μ are not independent electromagnetic constants. They are two faces of the charged rotors that already populate empty substrate (Section 3.1).

7.6.1 Charged rotor

A charged rotor is a compact Cosserat ring with inertia m and spring k , and it is a conversion cell of charge q in the sense used throughout these notes: a durable, steady-state converter between the two allotropes of the medium [36]. Those three numbers belong to the rotor species; they do not change from cell to cell.

7.6.2 Spring and ε

Spring $\leftrightarrow$ electrostatics. A potential V stretches the rotor by $\xi = qV/k$. The stored energy is

$$\frac{1}{2} k \xi^2 = \frac{1}{2} C V^2, \quad C = \frac{q^2}{k}. \quad (31)$$

C is the capacitance of one charged rotor. Packed at number density n ,

$$\varepsilon \sim nC = \frac{nq^2}{k}. \quad (32)$$

This is the same stretch that Section 7.2 packaged as $\mathbf{D} = \varepsilon \mathbf{E}$.

7.6.3 Governor, four populations, and μ

Rotor $\leftrightarrow$ magnetostatics. The same mass in motion is a current $I \sim qv$. The stored energy is

$$\frac{1}{2}mv^2 = \frac{1}{2}LI^2, \quad L = \frac{m}{q^2}. \quad (33)$$

L is the inductance of one charged rotor. Packed at the same density,

$$\mu \sim nL = \frac{nm}{q^2}. \quad (34)$$

This is the magnetic polarization of empty substrate, written as a Maxwell governor [3].

Empty substrate is globally neutral: a coarse cell contains equally many positive and negative rotors, with random orientations, so the mean conversion charge and the mean magnetic moment both vanish. An applied induction $\mathbf{B}$ loads the rotors by the torque on a current loop. Four populations, with $\mathbf{B}$ taken along $+z$:

conversion sign	ω versus $\mathbf{B}$	magnetic load
$q > 0$	parallel	up
$q > 0$	antiparallel	down
$q < 0$	parallel	down
$q < 0$	antiparallel	up

The two “up” populations find a new, higher equilibrium spin; the two “down” populations find a new, lower spin. They no longer cancel. The leftover is not given a separate magnetization field: it is already inside $\mathbf{B} = \mu \mathbf{H}$, with μ slaved to (ρ, V, Θ_{ij}) . $\mathbf{H}$ is the magnetic twin of $\mathbf{D}$: the auxiliary field whose source is free current.

7.6.4 Matching from L/C

Every factor of n sits in both ε and μ . Divide:

$$\frac{\mu}{\varepsilon} = \frac{L}{C} = \frac{mk}{q^4}. \quad (35)$$

The right-hand side is a property of one charged rotor. Crowding or rarefying copies cannot change it. Electromagnetic impedance matching

$$\frac{\mu}{\varepsilon} = Z_{0,\text{EM}}^2 = \frac{mk}{q^4} \quad (36)$$

is therefore derived, not imposed. The wave speed still feels the packing,

$$c_*^2 = \frac{1}{\varepsilon\mu} \propto \frac{1}{n^2}. \quad (37)$$

7.6.5 Map onto (ρ, V, Θ_{ij})

The dictionary then writes the same two numbers on the mechanical triple [34]. The acoustic tensor and the optical tensor are

$$A^{ij} = \frac{\Theta^{ij}}{\rho}, \quad n_{ij} = c_0 (A^{-1/2})_{ij}, \quad (38)$$

so that the principal speeds satisfy $c_k^2 = \Theta_k / \rho = c_0^2 / n_k^2$. The flow $\mathbf{V}$ is the same vector in the mechanical, optical and electromagnetic descriptions (the magnetoelectric / Fizeau slot). Once $Z_{0,EM}$ is the rotor ratio above, the principal permittivities and permeabilities are

$$\varepsilon_k = \frac{n_k}{c_0 Z_{0,EM}}, \quad \mu_k = \frac{n_k Z_{0,EM}}{c_0}, \quad (39)$$

hence $\mu = Z_{0,EM}^2 \varepsilon$. Thus ε , μ and $\mathbf{V}$ are a rewriting of (ρ, V, Θ_{ij}) , not a second ether. The derivation assumes a rigid rotor template: a well may change n , not the rotor's own (q, m, k) .

7.6.6 No gravitational analogue of the dipole sector

Gravity does not support a pair of opposite conversion regions of that kind. The seed T^{00} has one sign. There are no gravitational dipoles, bound or free. There is therefore no gravitational analogue of $\sigma \mathbf{E}$ built from opposite-mass pairs, and no gravitational analogue of a polarization current of such pairs. The time-dependent gravitational field is a modulation of Ψ itself (Section 8). Copying $\mathbf{J}_{\text{tot}} = \sigma \mathbf{E} + \varepsilon \partial_t \mathbf{E}$ onto gravity would invent a dipole sector that the source does not have.

8 Radiation

Electromagnetic and gravitational waves are not the same kind of motion of the sponge [4, 9, 8, 3, 14]. Whether they share one local speed is taken up in Section 8.4.

8.1 Electromagnetic waves

An electromagnetic wave is a dual excitation of the charged rotors on a frozen constitutive background (ρ, V, Θ_{ij}) . $\mathbf{D}$ is the charge-weighted positional displacement of the rotors from equilibrium. $\mathbf{H}$ is the charge-weighted spin density of the rotors relative to equilibrium.

Neighbouring rotors couple by mutual hydrodynamic induction, the same Biot–Savart / couple-stress coupling as in Section 3.1 [4, 7, 6]. A change of $\mathbf{D}$ in one cell torques the spin of the next cell; a change of $\mathbf{H}$ induces a change of $\mathbf{D}$. That first-order coupling is Faraday's law plus the Ampère–Maxwell law in empty substrate [3, 32],

$$\nabla \times \mathbf{E} = -\mu \partial_t \mathbf{H}, \quad \nabla \times \mathbf{H} = \partial_t \mathbf{D}, \quad (40)$$

or, when μ is uniform,

$$\nabla \times \mathbf{B} = \mu \partial_t \mathbf{D}. \quad (41)$$

The Maxwell pair is therefore the coarse writing of the rotor coupling, not an extra postulate. Units are SI: $\mathbf{D}$ in C m^{-2} , $\mathbf{H}$ in A m^{-1} , $\mathbf{B}$ in T, μ in H m^{-1} .

Electric charge has two signs [36]. The total charge of an isolated apparatus is conserved, so the monopole moment cannot oscillate. The lowest motion that changes the pattern of lines of the electric flux density without changing the total charge is an oscillation of two opposite conversion regions against each other. That is an electric dipole.

8.2 Gravitational waves

A gravitational wave is a quadrupolar, transverse, traceless modulation of the tensors ε and μ , equivalently of (ρ, V, Θ_{ij}) . It is a bulk modulation of the linear densities of the charged microvortices in the two directions perpendicular to the propagation direction [14]. It is not a second polarization of the electromagnetic wave.

Because ε and μ set the local wave speed and the local length and time standards of emergent matter (Section 9), an internal observer sees the modulation as a wiggle of spacetime itself.

The gravitational seed T^{00} has one sign. The total energy of an isolated source is conserved, so the gravitational monopole cannot oscillate. The gravitational dipole moment of an isolated source is its centre-of-energy. In the centre-of-energy frame that dipole does not oscillate; equivalently, conservation of momentum forbids dipole radiation. The lowest motion that changes the constitutive state without moving the total energy or the centre-of-energy is a time-dependent deformation of the source, $\delta\sigma^{ij}(t)$. That is a quadrupole [14]. The absence of gravitational dipole radiation is the same fact as the absence of gravitational dipoles in Section 7.

8.3 Retardation

The coupling of Section 8.1 is local: a rotor talks to its neighbours through the interstitial flow, not to the whole medium at once.

For an electromagnetic source a moving converter changes $\mathbf{D}$ and $\mathbf{H}$ at its own cell. Distant rotors still carry the old dual pattern. The inductive coupling restores them after the delay of the interstitial flow. That travelling mismatch is the electromagnetic wave.

For a gravitational source a moving mass changes the local linear density of rotors, hence ε and μ , and hence (ρ, V, Θ_{ij}) (Section 8.2). Distant cells still carry the old packing. The mechanical relaxation of that packing restores them after the same delay. That travelling mismatch is the gravitational wave.

8.4 One characteristic speed

The two channels are not on the same footing, and that is the principle of the argument.

For electromagnetism the closed first-order system is already on the page (Section 8.1). The fields $\mathbf{E}$, $\mathbf{B}$, $\mathbf{D}$ and $\mathbf{H}$ are coupled by Faraday and Ampère–Maxwell. The tensors ε and μ enter those equations as *coefficients*: they convert $\mathbf{E}$ into $\mathbf{D}$ and $\mathbf{H}$ into $\mathbf{B}$. They are not themselves the unknowns of that system. The wave speed of the pair then follows at once,

$$c_{\text{EM}}^2 = \frac{1}{\varepsilon\mu}. \quad (42)$$

Section 7.6 identifies the product with the packing of the charged rotors, and the dictionary writes

$$\frac{1}{\varepsilon\mu} = \frac{\Theta}{\rho} = c_*^2. \quad (43)$$

For gravity the unknowns *are* ε and μ (equivalently (ρ, V, Θ_{ij})). A gravitational wave is a transverse-traceless modulation of those tensors (Section 8.2). No elastic-like field equation for ε and μ has been written in these notes. Until that system exists one cannot derive c_G by the same first-order route that gives c_{EM} .

The argument that remains is only this. Whatever dynamics eventually closes ε and μ is a dynamics of the same rotors whose packing already defines those tensors. The only characteristic speed that packing has supplied is $c_* = \sqrt{\Theta/\rho} = 1/\sqrt{\varepsilon\mu}$. A completed elastic theory of ε and μ is therefore expected to propagate its small transverse-traceless disturbances on that same cone, not on a second one. That is a material principle and a programme, not a finished proof.

If the expectation holds, the retarded time of an electromagnetic dipole and the retarded time of a gravitational quadrupole, computed in the same region, are the same function of Ψ . Equal arrival of a light pulse and a gravitational wave is then one medium, not a coincidence of two theories.

9 Effective metric and local frame

Linear irreversible thermodynamics gives Ψ , Φ and $T + \theta$. It does not give geodesics. Localized energy concentrations move on an effective metric because they are excitations of the same medium whose constitutive state is Ψ . That map is a second constitutive law, already written in the dictionary [34].

From (ρ, V, Θ_{ij}) one reconstructs a refractive tensor n_{ij} and a flow V^i . The effective metric in laboratory coordinates is

$$\begin{aligned} g_{00} &= -\frac{1}{\bar{n}} + n_{ij}v^iv^j, \\ g_{0i} &= n_{ij}v^j, & v^i &= V^i/c_0, & \bar{n} &= (\det n)^{1/3}. \\ g_{ij} &= n_{ij}, \end{aligned} \tag{44}$$

This metric is the geometry measured by a local clock and three local rods made of the medium at that event. The clock rate is the characteristic period of the medium, $\tau_* \sim \ell_*/c_*$. The rods lie along the principal axes of n_{ij} (equivalently of Θ^{ij}). Those four instruments form a tetrad

$$\theta^0 = \bar{n}^{-1/2} dX^0, \quad \theta^a = (\sqrt{\bar{n}})^a_i (dx^i + v^i dX^0), \tag{45}$$

with $X^0 = c_0 t$. Then $g_{\mu\nu} = \eta_{AB} \theta^A_\mu \theta^B_\nu$. The shift $dx + V dt$ is the drag of the rods by the flow; it is the same V whose curl appears in Section 7.

The tetrad is not a fifth field. It is reconstructed from the values of (ρ, V, Θ_{ij}) at one event. The background coordinates remain Euclidean and Newtonian. What changes is the reading of those coordinates by instruments made of the medium.

To first order in Ψ/c^2 , $e^A_\mu = \delta^A_\mu + \Psi^A_\mu/c^2$, which is the tetrad of the perturbation $h_{\mu\nu} = 2\Psi_{\mu\nu}/c^2$. The Papapetrou representation and the optical metric therefore agree in the weak-field limit, provided the Newtonian sign $\delta\bar{n} = -2\Psi^{00}/c^2$ is used when Ψ^{00} is the Newtonian potential.

Impedance matching is a restriction on the allowed fields Ψ . It is not a field equation. It keeps electromagnetism and gravity on the same characteristic cone.

10 Open calculations

The note does not replace a calculation. The following items are well posed and have not been written.

- The elastic-like field equations for $\boldsymbol{\varepsilon}$ and $\boldsymbol{\mu}$ themselves (or for $(\delta\rho, \delta\Theta)$), and the characteristic speed of that system compared with $1/\sqrt{\varepsilon\mu}$ (Section 8.4).
- The first-order induction system for the pair $(\Phi, \nabla \times V, F^{ijk})$, with a single speed $c_*[\Psi]$.
- The Onsager–Casimir matrix [16, 20, 24], including the antisymmetric gyrotropic block.
- The identification $c_*^2 = \Theta/\rho$, with the electromagnetic and gravitational couplings written in units.
- The near-zone and wave-zone fields of an electric dipole and of a gravitational quadrupole, with the same retarded time.
- Whether t_P is primitive or equals $\ell_*/c_*(\Theta_0, \rho_0)$, with $\hbar = E_0/\omega_0$.
- A single Lagrangian that yields both the electromagnetic source and the tensor Gauss law.

These are calculations within the present note.

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